Formative 2 Report: Modeling Human Activity States Using Hidden Markov Models

Module: Machine Learning
Assessment: Formative 2 - Hidden Markov Models
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Repository: https://github.com/dmutoni/Hidden-Markov-Model

1. Background and Motivation

Human Activity Recognition (HAR) plays an important role in modern intelligent systems such as wearable health monitors, rehabilitation tracking systems, smart homes, and assisted living technologies. In this project, we focused on a lightweight home rehabilitation monitoring use case. The goal is to provide a low-cost approach for recognizing simple human activities using only the sensors available in a smartphone.

Our system detects whether a user is standing, walking, jumping, or still using accelerometer and gyroscope signals collected from a smartphone. These sensors generate noisy measurements, meaning the real activity state is not directly observable. Hidden Markov Models (HMMs) are particularly suitable for this type of problem because they model temporal dependencies. The current activity state depends on the previous state, while the observed sensor features are probabilistic emissions from the hidden activity state.

2. Data Collection and Preprocessing

2.1 Data Collection Setup

We collected smartphone motion recordings using accelerometer and gyroscope sensors with timestamped CSV logs. The following activities were recorded: Standing, Walking, Jumping, and Still (phone placed on a flat surface).

Dataset summary:
• Total recordings: 51
• Jumping files: 12
• Standing files: 13
• Still files: 13

• Walking files: 13

Total duration per activity:
• Jumping: 91.095 seconds
• Standing: 125.274 seconds
• Still: 101.415 seconds
• Walking: 122.640 seconds

This satisfies the requirement that each activity contains at least 1 minute and 30 seconds of recorded data.

2.2 Device and Sampling Rate

Two smartphones were used during data collection:
• iPhone 15 – used for Jumping and Still recordings
• SM-S918U1 – used for Standing and Walking recordings

Since different devices may record sensor data at slightly different sampling rates, we harmonized all recordings by resampling the data to a common frequency of 50 Hz. This ensures that all recordings produce comparable feature windows regardless of the device used.

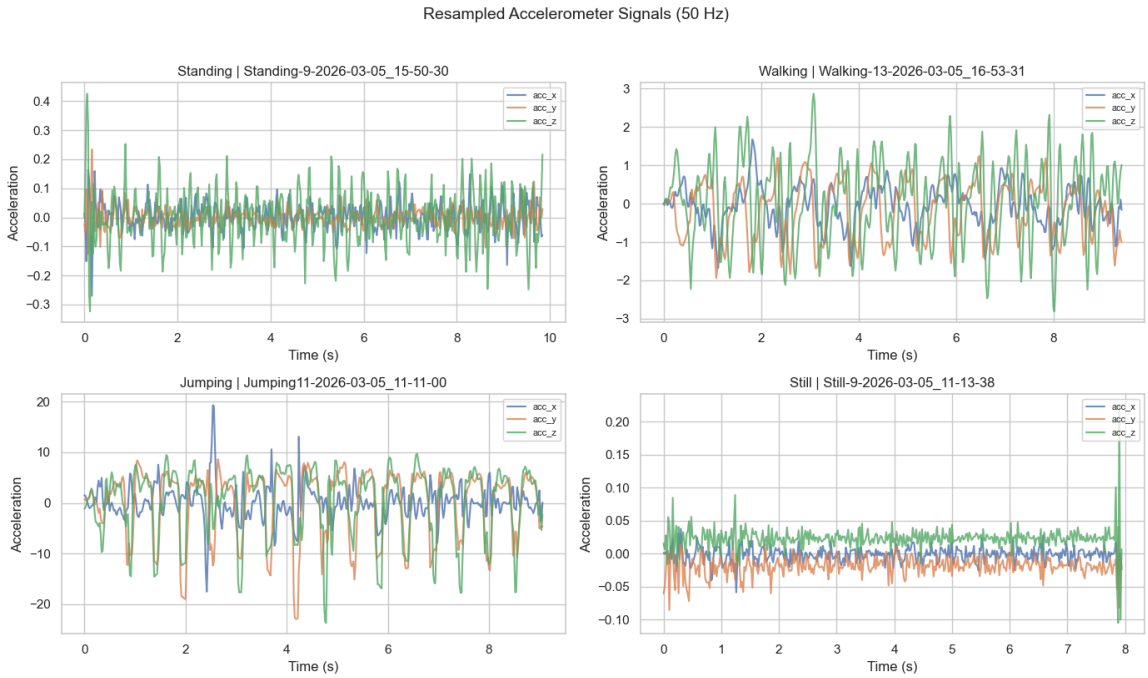
2.3 Data Cleaning and Synchronization

The preprocessing pipeline included the following steps:
1. Load accelerometer and gyroscope streams.
2. Align sensor rows using nearest timestamps.
3. Combine the sensors into a synchronized multivariate signal.
4. Resample all signals to a uniform 50 Hz timeline.
5. Remove incomplete or corrupted windows.

2.4 Windowing Strategy

Window size: 2 seconds
Sampling rate: 50 Hz
Samples per window: 100
Overlap: 50%

A 2-second window was chosen because it captures periodic movement patterns such as walking or jumping cycles while still maintaining good temporal resolution. The 50% overlap allows smoother transitions between windows without significantly increasing computational cost.



3. Feature Extraction

3.1 Time-Domain Features

The following time-domain features were extracted from each window:

- Mean
- Standard deviation
- Variance
- Root Mean Square (RMS)
- Signal Magnitude Area (SMA)
- Axis correlations
- Magnitude range
- Jerk RMS
- Low-motion ratio

These features help separate high-motion activities such as jumping from low-motion activities such as standing or still.

3.2 Frequency-Domain Features

Frequency-based features were derived using Fast Fourier Transform (FFT):

- Dominant frequency
- Spectral energy in the 0.5–3 Hz band

Periodic activities like walking and jumping show strong peaks in this frequency band, which helps distinguish them from static activities.

3.3 Normalization

All features were normalized using Z-score normalization computed from the training dataset. This ensures that features with larger scales do not dominate the model during training.

4. Hidden Markov Model Setup

The Hidden Markov Model consists of the following components:

Hidden States (Z): latent activity states
Observations (X): feature vectors extracted from each window
Transition Matrix (A): probabilities of moving between states
Emission Probabilities (B): Gaussian distributions over features
Initial State Probabilities (π): probability of starting in each state

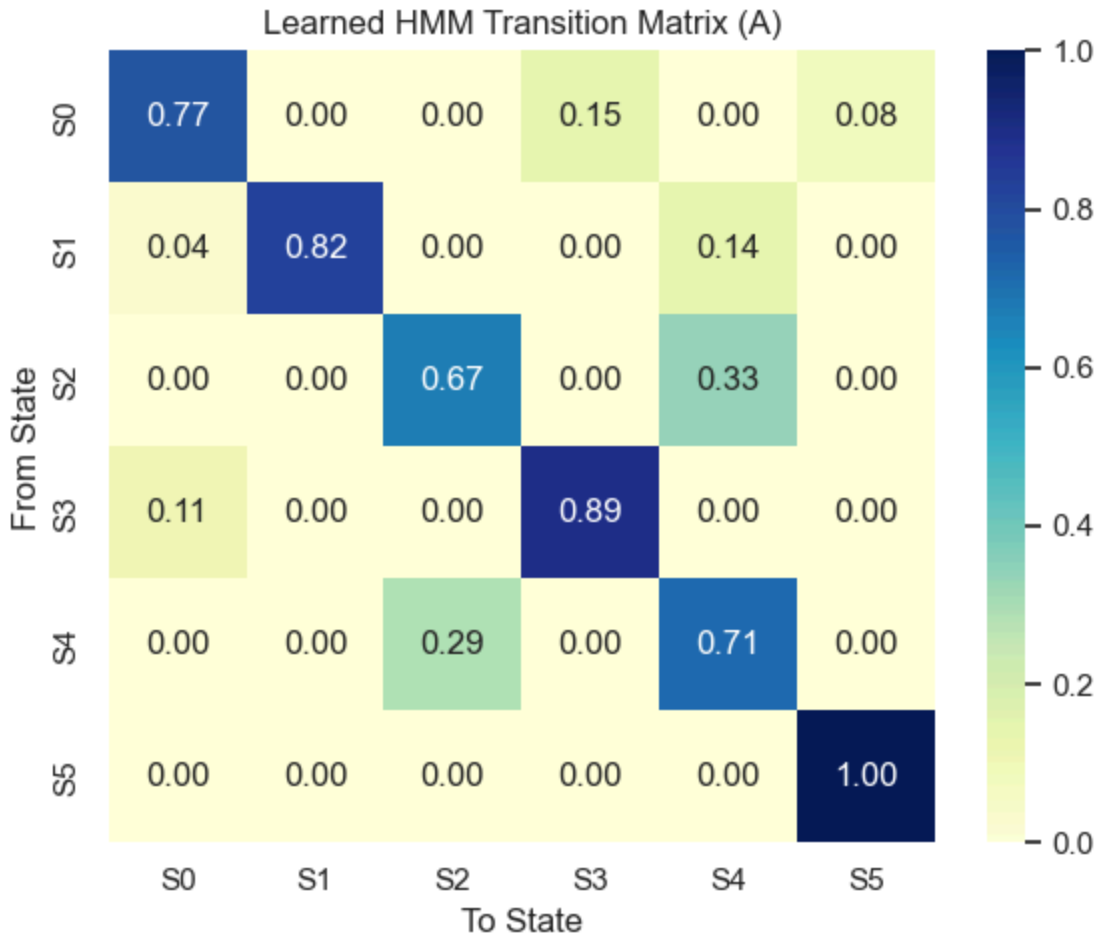
The model was implemented using the hmmlearn library with 6 latent states, allowing the model to capture sub-variations within activities.

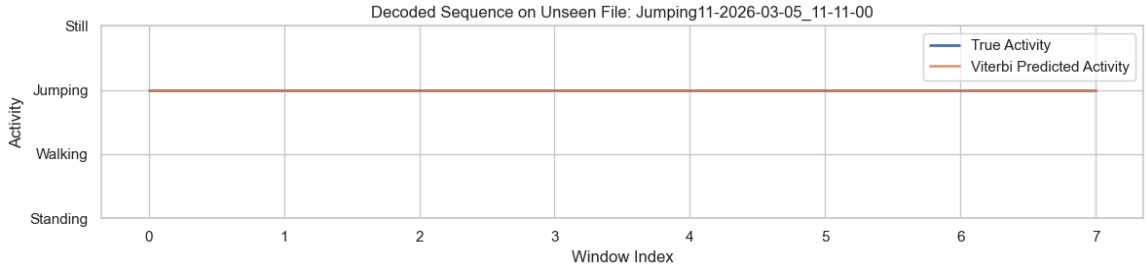
Training (Baum–Welch)

The Baum–Welch algorithm was used to estimate the model parameters using Expectation-Maximization. The GaussianHMM implementation from hmmlearn was used with diagonal covariance matrices.

Decoding (Viterbi)

The Viterbi algorithm was used to decode the most likely sequence of hidden states for unseen test observations.



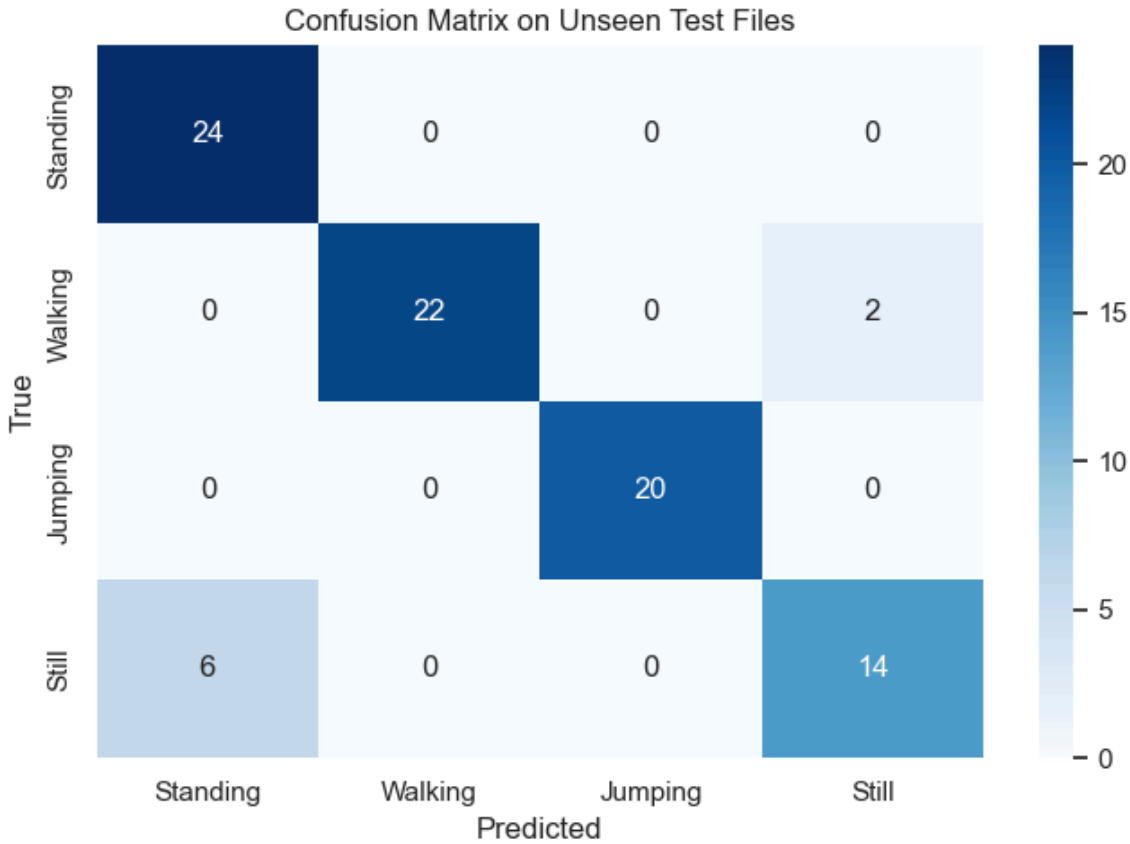


5. Results and Evaluation

Evaluation was performed using unseen test files that were not used during training. The model demonstrated strong generalization performance.

Activity	Number of Samples	Sensitivity	Specificity	Overall Accuracy
Standing	24	1.00	0.91	0.93

Walking	24	0.92	1.00	0.98
Jumping	20	1.00	1.00	1.00
Still	20	0.70	0.97	0.91



6. Analysis and Reflection

Jumping was the easiest activity to detect because it produces strong periodic motion patterns. Standing was also easy due to its consistent variance profile. The hardest distinction was between Still and Standing because small body movements while standing can resemble still conditions.

The transition matrix also revealed realistic temporal behavior, where activities tend to persist across consecutive windows rather than switching randomly.

7. Collaboration and GitHub Contribution

Team Member	Responsibilities	Evidence

Mutoni Uwingeneye Denyse	HMM implementation, evaluation and documentation	Notebook commits and commits to documentation and report
Oreste Abizera	Feature engineering, HMM implementation, evaluation	Notebook commits and repository structure

8. Conclusion

This project demonstrates that smartphone sensor data combined with Hidden Markov Models can effectively recognize basic human activities. The final model achieved over 90% accuracy on unseen data, showing strong generalization performance. Future work could include collecting more data, adding additional sensors, and experimenting with hybrid models.